HTDC: Hyperbolic Transformer With Dual-Momentum Contrastive Learning for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) remains a challenging task due to the complex spectral variability and spatial heterogeneity of real-world scenes. Reconstruction-based methods have been widely explored for deep learning-based HAD, but they rely on the assumption that background pixels can be faithfully reconstructed while anomalous pixels cannot. Although intuitive, this dichotomy oversimplifies the spectral–spatial characteristics of hyperspectral data and often fails in practice. In particular, when background regions exhibit strong variability or when anomalies share spectral similarities with their surroundings, reconstruction-based methods tend to generate false alarms and suffer from limited robustness. To overcome these limitations, this article moves beyond the reconstruction-based paradigm and proposes HTDC, a novel hyperbolic Transformer with a dual-momentum contrastive learning (DMCL) framework, which shifts the focus from background reconstruction to discriminative representation learning at both global and local levels. First, a custom-designed multiview data augmentation module generates complementary views from different perspectives, enriching spectral diversity and alleviating the limitations of contrastive learning (CL) under homogeneous samples. Second, a hyperbolic Transformer encoder exploits the hierarchical and structured nature of hyperspectral data, where hyperbolic geometry enables more effective attention aggregation than Euclidean space. Subsequently, a dual-momentum mechanism with dual temporal queues further stabilizes global learning while dynamically adapting to local changes, thereby mitigating sensitivity to class imbalance and spectral drift. Finally, in the detection stage, a sliding dual-window detection strategy with exponential mapping and guided filtering is employed to suppress background interference and enhance spatial contrast. Extensive experiments on multiple real hyperspectral images acquired by different sensors across various scenes demonstrate that the proposed HTDC consistently outperforms the state-of-the-art methods in HAD tasks.**

***Index Terms*—Contrastive learning (CL), dual-momentum, hyperbolic transformer, hyperspectral anomaly detection (HAD), multiview data augmentation.**

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I. Introduction

Hyperspectral imaging (HSI) acquires detailed spectral information across tens to hundreds of contiguous narrow bands, enabling the generation of high-dimensional spectral response vectors for each pixel while preserving spatial structures [1], [2]. Due to this unique capability, HSI has been extensively employed in a wide range of remote sensing tasks, including classification [3], [4], [5], anomaly detection [6], [7], [8], target detection [9], [10], [11], and band selection [12], [13]. From a perspective of material identification, the fine spectral resolution of HSI enables precise discrimination of subtle compositional variations, which is particularly advantageous for detecting anomalous targets in complex backgrounds such as camouflage detection and pollutant monitoring. In general, anomalies in HSIs exhibit three characteristics: 1) low probability of occurrence; 2) small spatial extent; and 3) significant spectral differences from the surrounding background [14], [15]. Existing hyperspectral anomaly detection (HAD) methods can be broadly classified into three categories: statistical-based methods, representation-based methods, and deep learning-based methods, each of which has produced representative approaches as outlined below.

Statistical-based methods assume that background pixels follow a specific distribution, with anomalies identified as outliers. One of the most representative approaches is the RX detector [16], which models the background pixels using a multivariate Gaussian distribution and detects anomalies as pixels that exhibit significant deviations from this distribution. However, in real-world scenarios, the background is often too complex to be accurately represented by a single multivariate Gaussian model, resulting in degraded detection performance. Various such as local RX (LRX) [17], weighted RX (WRX) [18], kernel-based RX (KRX) [19], and recursive RX with multiattribute profiles [20] attempt to improve the robustness of the conventional RX detector, but all remain fundamentally constrained by their reliance on predefined distributional assumptions.

To avoid reliance on inaccurate distribution assumptions, representation-based methods have attracted significant attention [21]. These approaches represent hyperspectral data as linear or nonlinear combinations of a set of bases or dictionaries. Main approaches in this category include low-rank representation (LRR), sparse representation, collaborative representation, and tensor-based modeling. For instance, the collaborative representation-based detector (CRD) [22] is designed to exploit the sparsity and locality characteristics of HSI by assuming that background pixels can be linearly represented by their neighboring pixels, while anomalies cannot. However, the performance of CRD is sensitive to the selection of dual-window sizes across different datasets. LRR-based methods model hyperspectral data as a linear combination of a small number of intrinsic components, reconstruct the background, and detect anomalies as reconstruction residuals. While effective in capturing global structures, LRR tends to ignore local structural information. To address this limitation, GTVLRR incorporates graph regularization and total variation regularization into the LRR model [23], preserving global structural characteristics while simultaneously capturing local geometric properties. Given the widespread nature of background distributions in HSI and the rarity of anomalies, hybrid methods combining low-rank and sparse constraints have been explored to refine anomaly-background separation. Representative methods include low-rank and sparse representation (LRASR) [24] and low-rank matrix decomposition using Mahalanobis distance (LSMAD) [25].

In practical applications, both statistical-based and representation-based methods encounter significant challenges when dealing with the complex and high-dimensional nature of hyperspectral backgrounds. To address these limitations, deep learning has been increasingly explored in recent years [26], [27], owing to its ability to model nonlinear spectral–spatial features. Current deep learning-based HAD methods primarily focus on three types of architectures: convolutional neural networks (CNNs), generative adversarial networks (GANs), and autoencoders (AEs). CNN-based methods are generally unsuitable for practical anomaly detection tasks due to their reliance on prior labeled information for supervised training [28], [29]. To overcome this, DeCNN-AD [30] introduces a plug-and-play CNN denoiser to regularize representation coefficients, enabling unsupervised learning within a traditional framework. Furthermore, to address the computational burden of retraining and the constraints of fixed input dimensions often faced by CNNs, AdaptHAD [31] proposes an adaptive hybrid network that integrates a CNN with a Transformer to simultaneously extract local and global features. GANs enhance the generalization ability of neural networks through adversarial learning [32], [33]. Although their effectiveness in HAD has been demonstrated, many GAN-based variants suffer from unstable training and interference from redundant information, leading to high false-alarm rates and reduced detection accuracy. To mitigate this, CL-BioGAN [8] incorporates biologically inspired synaptic plasticity to actively balance memory stability and learning flexibility, ensuring robust detection in cross-domain scenarios. In contrast, AE-based methods have been increasingly adopted in HAD owing to their capacity to extract deep features with nonlinear characteristics and their strong unsupervised learning capabilities [34]. For instance, RGAE [35] integrates graph constraints and superpixel segmentation, while Auto-AD [36] employs adaptive loss adjustment to avoid manual parameter tuning. However, AE-based methods may inadvertently reconstruct anomalies along with the background, thereby degrading detection performance. GAED [37] addresses this by incorporating a feedback strategy to suppress the representation of anomaly features. Similarly, to prevent anomaly leakage, a blind-spot network [38] utilizing multiscale mask convolutions has been proposed, while FCAE-DCAC [39] integrates dual clustering priors with latent feature adversarial consistency to ensure prioritized background reconstruction. Other approaches embed physical priors, such as TAEF [40], which integrates an extended multilinear mixing model and employs a Transformer for improved performance. In order to exploit the spatial–spectral characteristics inherent in HSI, S3S-LTGRMA [41] integrates spectral, spatial, and structural features through superpixel segmentation and a Transformer backbone. Beyond Transformers, emerging architectures have also been explored; MMR-HAD [42] leverages the Mamba model with random masking to inhibit anomaly reconstruction, while DWSDiff [43] introduces a dual-window spectral diffusion model for precise background estimation. Despite these advances, most AE-based methods remain fundamentally tied to the reconstruction assumption—that background pixels can be accurately reconstructed while anomalies cannot. In practice, anomalies may be partially reconstructed with the background, causing distortion and reducing robustness. This problem is exacerbated in high-dimensional data with complex land-cover distributions, where the binary distinction of "background versus anomaly" often breaks down.

To overcome these limitations, this article moves beyond the reconstruction paradigm and adopts contrastive learning (CL) [44], [45], [46] to develop a discriminative anomaly detection approach driven by intersample differences. Building upon this, this article proposes a hyperbolic transformer with dual-momentum contrastive learning (HTDC), a unified framework that synergistically integrates multiview data augmentation, a hyperbolic Transformer encoder, and a dual-momentum mechanism to enable HAD through CL. In particular, a multiview data augmentation strategy is employed to generate semantically consistent yet diverse views, providing the foundation for robust contrastive representation learning. A Transformer equipped with hyperbolic attention is then constructed to capture the hierarchical structural dependencies inherent in HSI. During the CL, a dual-momentum strategy with dynamic queues is introduced. The high-momentum encoder, updated slowly with a large momentum coefficient, captures global contextual semantics, whereas the low-momentum encoder, updated more frequently with a smaller momentum, focuses on local, fine-grained spectral representations. In this process, a spectral feature encoder serves as the query branch, performing CL by comparing its outputs with those from both momentum branches. Finally, in the detection stage, a sliding dual-window strategy is applied, combined with a guided filter and a power function, to effectively suppress background interference and enhance spatial–spectral contrast. The main contributions of this work are summarized as follows.

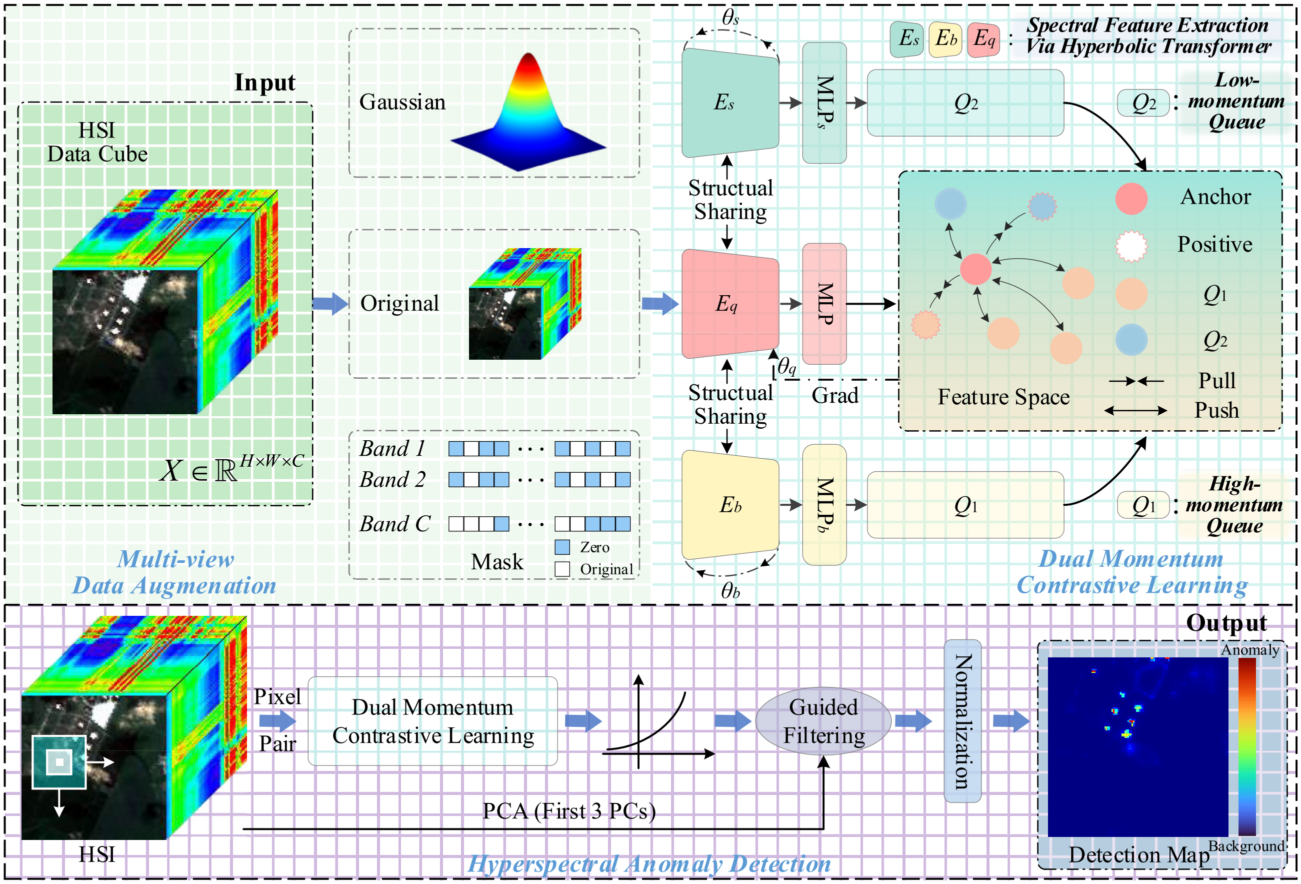

Fig. 1. Flowchart of the proposed HTDC method.

1) *Multiview Augmentation for CL:* A strategy that generates diverse perspectives by simulating data completeness, resolution variation, and band absence. Through Gaussian noise perturbation and random spectral band masking, it enriches spectral diversity and alleviates the limitations of CL under homogeneous samples.

2) *Hyperbolic Transformer Encoder:* An architecture built upon the conventional Transformer, incorporating a hyperbolic-optimized multihead self-attention to exploit the hierarchical structure of hyperspectral data. By modeling feature interactions in hyperbolic space, it enhances the representation of complex hierarchical dependencies.

3) *DMCL:* A spectral discriminative model fundamentally different from reconstruction-based approaches, incorporating a dual-momentum mechanism with two temporal queues to stabilize global optimization while adapting to local variations, thereby mitigating sensitivity to class imbalance and spectral drift.

The remainder of this article is organized as follows. Section II presents the details of the proposed HTDC. Section III provides experimental results and analysis to validate the effectiveness of the proposed approach. Finally, conclusions are drawn in Section IV.

II. Methodology

This section presents a comprehensive overview of the proposed hyperbolic transformer with dual-momentum contrastive learning (HTDC) framework for HAD. The framework consists of five main components: a multiview data augmentation to enhance the quality and diversity of data samples, a spectral feature extraction network based on a hyperbolic Transformer to capture hierarchical spectral–spatial dependencies, a CL mechanism with a dual-momentum strategy to improve feature discrimination, and a dual-window anomaly detection module that applies the proposed model with background suppression. The overall architecture of the proposed HTDC is illustrated in Fig. 1.

A. Multiview Data Augmentation

CL typically begins with data augmentation to increase data variability and provide diverse views of the same instance. Conventional augmentation techniques in computer vision—such as cropping, flipping, rotation, and color transformations—are not well-suited for hyperspectral remote sensing, particularly in HAD. The heavy reliance on spectral information makes these conventional methods inadequate for fully exploiting HSI characteristics. To address this limitation, a multiview data augmentation strategy is proposed that considers three key aspects: spectral completeness, resolution variation, and spectral absence, enriching the diversity of data distribution within the CL framework, as illustrated in Fig. 1.

For spectral completeness, the original data is first preserved without any augmentation to retain its full spectral information. To simulate resolution variation, Gaussian noise

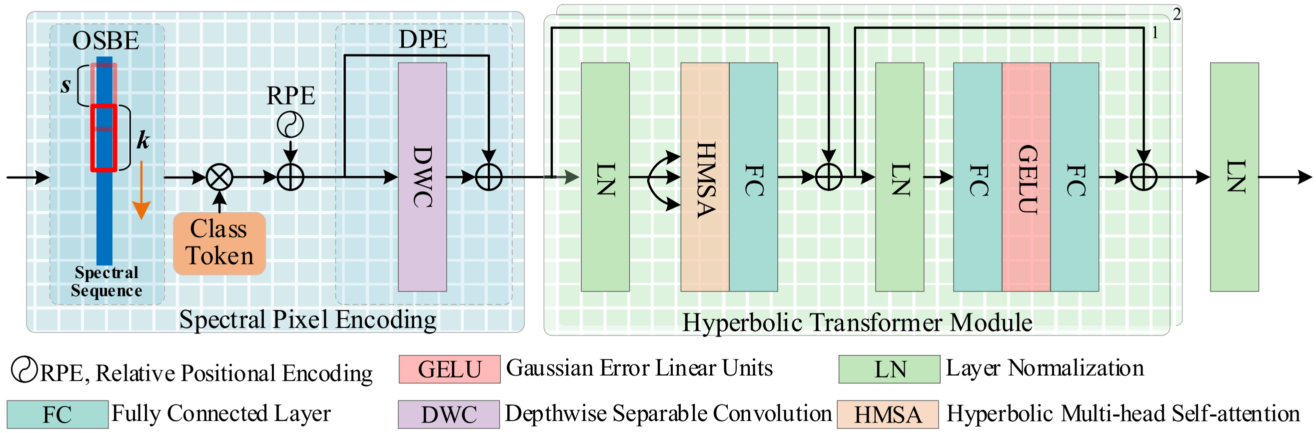

Fig. 2. Spectral feature extraction via hyperbolic transformer.

is then introduced for the generation of positive and negative sample pairs used in training [47], [48]. Zero-mean Gaussian noise introduces mild perturbations to the original high-frequency features, which, in moderate amounts, can enhance the network's learning capability, expressed as follows:

$$X' = X + \beta \times N \tag{1}$$

where X denotes the original input, N is standard Gaussian noise, and β is set to 0.005.

Finally, spectral absence is simulated through random band-wise masking. In particular, for each spectral band, a random binary mask is applied to 20% of the pixel values, generating diverse spectral views. This process emulates partial spectral loss and encourages the model to learn robust feature representations under missing-band conditions.

B. Spectral Feature Extraction via Hyperbolic Transformer

This section is motivated by two key considerations. First, constructing a fine-grained spectral feature extraction module is essential for improving the effectiveness of CL. Second, hyperspectral data naturally exhibit hierarchical spectral structures: local variations between adjacent bands, holistic spectral signatures within individual pixels, and statistical dependencies across the dataset. To model such structures, a hyperbolic space is introduced, whose non-Euclidean properties are particularly well-suited for representing hierarchical data distributions. Integrating this geometric framework with the Transformer architecture enhances the model's capacity to represent structured spectral information, as illustrated in Fig. 2. The spectral feature extraction process consists of two components: spectral pixel encoding, which performs initial spectral feature extraction, and the hyperbolic Transformer module, which models hierarchical dependencies in a hyperbolic latent space.

1) Spectral Pixel Encoding: The first component, spectral pixel encoding, generates high-quality spectral token sequences to support efficient and fine-grained feature extraction in the hyperbolic Transformer. Considering the inherent correlations among hyperspectral bands, an Overlapping Spectral Block Embedding (OSBE) module is first employed to construct these tokens by controlling the overlap between adjacent spectral blocks. This design preserves sufficient spectral information within each token and enhances the Transformer encoder's ability to capture contextual dependencies. In particular, given an input HSI with C spectral bands, a 1-D convolution layer with a kernel size of k is applied. The stride parameter s ($s < k$) controls the degree of overlap between spectral blocks, with the overlap v defined as

$$v = k - s. \tag{2}$$

In addition, to address the inherent position-invariance of the standard Transformer self-attention mechanism, both relative positional encoding (RPE) and dynamic positional encoding (DPE) are incorporated into the architecture. The RPE module captures relative positional relationships among spectral bands, enhancing the Transformer's capacity to model sequential dependencies. The encoding is defined as:

$$\mathrm{RPE}\,(pos, 2i) = \sin\left(\frac{pos}{100000^{2i/d}}\right) \tag{3}$$

$$\mathrm{RPE}\,(pos, 2i+1) = \cos\left(\frac{pos}{100000^{2i/d}}\right) \tag{4}$$

where pos denotes the position of the token, i represents the dimension index, and d indicates the embedding dimension.

In contrast, DPE takes into account the strong local correlations of spectral features by integrating a depthwise separable convolution (DWC) layer with a kernel size of $k = 3$, as can be shown in the left side of Fig. 2.

2) Hyperbolic Transformer Module: While most existing Transformer-based methods operate in Euclidean space, their ability to represent the spatial hierarchies of HSI is limited. Considering that real-world data commonly exhibit hierarchical structures, this article incorporates the geometric properties of hyperbolic space into the Transformer architecture [49], [50], [51], [52], [53], [54]. This integration enables more effective modeling of hierarchical dependencies while maintaining the global context modeling capability of the Transformer, as illustrated on the right side of Fig. 2, along with the design of the Hyperbolic multihead self-attention (HMSA) module shown in Fig. 3.

The hyperbolic Transformer module is built upon the conventional Transformer framework. In the HMSA mechanism,

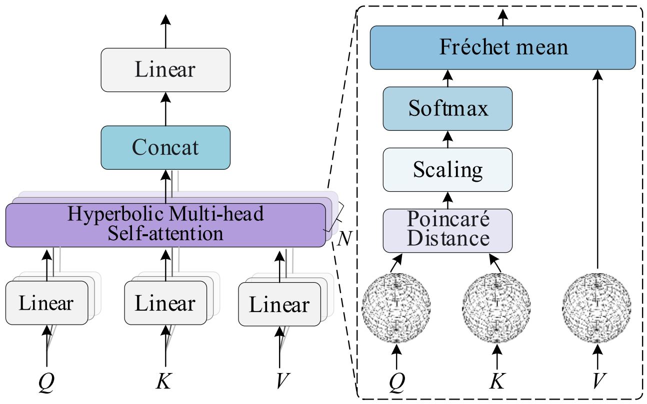

Fig. 3. HMSA module.

the input matrix X is linearly projected to obtain the query (Q), key (K), and value (V), as defined below

$$Q = W^q X,\ \ K = W^k X,\ \ V = W^v X \tag{5}$$

where W^q, W^k, and W^v are the corresponding weight matrices

To embed HMSA in hyperbolic space, a Poincaré projection is applied, mapping Q, K, and V onto a hyperbolic ball of radius r, as formulated in (6) and (7). This guarantees that all computations are performed within hyperbolic space, enabling more faithful modeling of hierarchical structures compared to Euclidean formulations.

$$proj(x) = x \cdot \min\left(1, \frac{r \cdot (1-\varepsilon)}{\|x\| + \varepsilon}\right) \tag{6}$$

$$Q' = proj(Q), K' = proj(K), V' = proj(V) \tag{7}$$

where r denotes a learnable radius parameter.

After projection, the attention weights are computed by measuring the hyperbolic distance between Q and K under the Poincaré ball model, as defined in the following equation:

$$d\left(Q', K'\right) = \frac{2}{\sqrt{c}} \cosh^{-1}\left(1 + \frac{2c \cdot \|Q' - K'\|_2^2}{\left(1 - c\|Q'\|_2^2\right)\left(1 - c\|K'\|_2^2\right) + \varepsilon}\right) \tag{8}$$

$$A' = \text{softmax}\left(-\frac{d\left(Q', K'\right)}{\sqrt{d_k}}\right) \tag{9}$$

where $c = 1/r^2$, ε is a small constant introduced for numerical stability, and d_k is a scaling factor used to prevent the attention weights from becoming excessively large, thereby ensuring the stability of the computed values.

Because Euclidean weighted averaging is not directly applicable in hyperbolic space, a hyperbolic aggregation strategy based on the Fréchet mean is adopted. In particular, the Fréchet mean is approximated by first performing a logarithmic map to project the points onto the tangent space, applying weighted averaging in the Euclidean domain, and then using an exponential map to project the result back into the hyperbolic space. To ensure numerical stability and geometric validity, a Poincaré projection is introduced throughout the process to constrain all results within the unit ball, as demonstrated in (10)–(15). Furthermore, given the rapid convergence speed of the Fréchet mean estimation, the number of iterations is fixed at $t = 2$, balancing computational efficiency and aggregation stability.

- Initialization

$$u_1 = proj\left(A' \cdot V'\right). \tag{10}$$

- Iterative refinement

$$u_t = \exp^c_{u_{t-1}}\left(A' \cdot \log^c_{u_{t-1}}\left(V'\right)\right) \tag{11}$$

$$A = proj\left(u_t\right) \tag{12}$$

where $proj(\cdot)$ denotes the Poincaré projection operator, and t refers to the aggregation iteration step.

- Exponential and logarithmic maps

$$\exp^c_x(v) = x \oplus_c \left(\tanh\left(\sqrt{c}\frac{\lambda^c_x\|v\|}{2}\right)\frac{v}{\sqrt{c}\|v\|}\right) \tag{13}$$

$$\log^c_x(y) = \frac{2}{\sqrt{c}\lambda^c_x} \cdot \tanh^{-1}\left(\sqrt{c}\|-x \oplus_c y\|\right) \cdot \frac{-x \oplus_c y}{\|-x \oplus_c y\|} \tag{14}$$

where $\lambda^c_x = 2/(1 - c\|x\|^2)$ is the conformal factor, and $\oplus_c$ denotes the Möbius addition in the Poincaré ball in the following equation:

$$x \oplus_c y = \frac{\left(1 + 2c\langle x, y\rangle + c\|y\|^2\right)x + \left(1 - c\|x\|^2\right)y}{1 + 2c\langle x, y\rangle + c^2\|x\|^2\|y\|^2} \tag{15}$$

where $\langle\cdot,\cdot\rangle$ denotes the inner product, $c = 1/r^2$, and r denotes a learnable radius parameter.

Finally, the outputs of all attention heads are concatenated to form the final output of the HMSA module, shown as follows:

$$\text{HMSA}(X) = \text{Concat}\left(A_1, A_2, \ldots, A_N\right) \tag{16}$$

where N denotes the number of attention heads.

Subsequently, this concatenated feature is fed into a feed-forward network. The feed-forward network in the Hyperbolic Transformer encoder consists of two linear layers with a Gaussian error linear units (GELUs) activation in between, as shown in Fig. 2.

Following the feed-forward stage, a layer normalization (LN) operation is applied to the output of the Transformer encoder. This standardizes the feature distribution, mitigating the internal covariate shift and facilitating smoother convergence for the subsequent contrastive mapping.

C. Dual-Momentum CL

To enable fine-grained CL in HAD tasks, this article moves beyond the conventional background anomaly dichotomy and focuses instead on capturing structural differences in spectral features. To this end, a DMCL framework is proposed to support hierarchical feature comparison across multiple scales. In particular, two encoders with distinct momentum coefficients are employed to maintain dual temporal queues: the high-momentum branch captures stable semantic structures, while the low-momentum branch enhances responsiveness to fine-grained spectral anomalies. In addition, a multiview data augmentation strategy is applied to enrich the diversity and informativeness of sample pairs. This design facilitates a more nuanced and hierarchical modeling of contrastive features.

First, a small batch of spectral pixels X_q is randomly sampled from the original HSI $X \in \mathbb{R}^{H\times W\times C}$. After applying multiview data augmentation, three augmented views are generated: X_q, X_b, and X_s, as defined in (17). Spectral pixels at the same spatial location across different views are considered as positive pairs, while those at different locations are treated as negative pairs

$$\begin{cases} X_q = \left[x_q^1; x_q^2; \cdots; x_q^N\right] \in \mathbb{R}^{N\times B} \\ X_b = \left[x_b^1; x_b^2; \cdots; x_b^N\right] \in \mathbb{R}^{N\times B} \\ X_s = \left[x_s^1; x_s^2; \cdots; x_s^N\right] \in \mathbb{R}^{N\times B} \end{cases} \tag{17}$$

where X_q denotes the original spectral data, X_b is produced via random band masking, and X_s is obtained by adding Gaussian noise.

The query encoder E_q is then applied to X_q, followed by a projection head MLP_q to obtain the feature matrix U_q. Simultaneously, the high-momentum encoder E_b and the low-momentum encoder E_s extract spectral features from X_b and X_s, respectively. These features are then passed through corresponding projection heads, MLP_b and MLP_s, to produce feature matrices U_b and U_s, which are stored in two temporally updated queues for subsequent CL, as expressed in the following equation:

$$\begin{cases} \mathrm{U}_q = \left[u_q^1; u_q^2; \cdots; u_q^N\right] \in \mathbb{R}^{N\times d_{\mathrm{MLP}_q}} \\ \mathrm{U}_b = \left[u_b^1; u_b^2; \cdots; u_b^N\right] \in \mathbb{R}^{N\times d_{\mathrm{MLP}_b}} \\ \mathrm{U}_s = \left[u_s^1; u_s^2; \cdots; u_s^N\right] \in \mathbb{R}^{N\times d_{\mathrm{MLP}_s}}. \end{cases} \tag{18}$$

Finally, the feature U_q is jointly compared with the features U_b and U_s, which are stored in the high-momentum queue Q_1 and the low-momentum queue Q_2, respectively. These features are fed into a contrastive loss function designed to maximize the similarity between positive pairs while minimizing the similarity between negative pairs, thereby enforcing the encoder to learn discriminative spectral representations. In this work, the similarity between feature pairs is measured using the dot product. The proposed dual-momentum contrastive loss adopts the InfoNCE formulation [55], which is defined as follows:

$$l_b = \frac{1}{N}\sum_{i=1}^{N} -\log \frac{\exp\left(u_q^i \cdot u_b^i/\tau_1\right)}{\sum_{j=0}^{Q_1}\exp\left(u_q^i \cdot u_b^j/\tau_1\right)} \tag{19}$$

$$l_s = \frac{1}{N}\sum_{i=1}^{N} -\log \frac{\exp\left(u_q^i \cdot u_s^i/\tau_2\right)}{\sum_{j=0}^{Q_2}\exp\left(u_q^i \cdot u_s^j/\tau_2\right)} \tag{20}$$

$$l = \lambda \cdot l_b + (1-\lambda)\cdot l_s \tag{21}$$

where λ serves as a balancing parameter that controls the relative contributions of two branches, while τ_1 and τ_2 denote the temperature parameters associated with the two momentum queues, respectively. Each InfoNCE loss term consists of one positive sample embedding (anchor) and a set of negative sample embeddings drawn from the queues Q_1 and Q_2.

During training, the consistency of the dual memory queues is maintained via momentum update mechanisms. To balance the learning of global and local spectral features.

1) The high-momentum encoder E_b employs a larger momentum coefficient n, resulting in slower parameter updates that preserve the long-term statistical characteristics of HSI. A larger queue, Q_1, is also maintained to store extensive historical negatives, enhancing the learning of global spectral invariance.

2) The low-momentum encoder E_s, in contrast, uses a smaller momentum coefficient m, enabling faster updates that improve sensitivity to local spectral variations. A smaller negative sample queue Q_2 is also adopted to reduce the impact of outdated negative samples on local discriminative learning. The overall parameter update process can be formally defined as follows:

$$\begin{cases} \theta_b \leftarrow n\theta_b + (1-n)\,\theta_q \\ \theta_{b\text{-MLP}} \leftarrow n\theta_{b\text{-MLP}} + (1-n)\,\theta_{q\text{-MLP}} \\ \theta_s \leftarrow m\theta_s + (1-m)\,\theta_q \\ \theta_{s\text{-MLP}} \leftarrow m\theta_{s\text{-MLP}} + (1-m)\,\theta_{q\text{-MLP}} \end{cases} \tag{22}$$

where θ_q, θ_b, and θ_s denote the parameters of the encoders E_q, E_b, and E_s, respectively, while $\theta_{q-\mathrm{MLP}}$, $\theta_{b-\mathrm{MLP}}$, and $\theta_{s-\mathrm{MLP}}$ correspond to the parameters of the projection heads MLP_q, MLP_b, and MLP_s, respectively. In contrast, the query encoder E_q and its projection head MLP_q are updated through gradient backpropagation. An overview of the relationships among these parameters is provided in Fig. 1.

D. Anomaly Detection

In the detection phase, the original HSI is fed into the well-trained model to effectively identify anomalies in the feature space. Owing to its self-supervised training scheme, the proposed method functions as a reliable discriminative detector. In the learned feature space, the anomalous targets exhibit distinctive characteristics compared to their surrounding backgrounds.

1) Sliding Dual-Window Anomaly Detection: An anomaly detection strategy based on a sliding dual window is adopted. The inner and outer window sizes of the dual window are set to 3 and 5, respectively. Given an input HSI, each pixel to be tested is paired with its 16 surrounding pixels within the outer window to form multiple pixel pairs. The model computes similarity scores for each pair, and the average of these scores is taken as the final output. If the score exceeds a predefined threshold η, the pixel is classified as an anomaly; otherwise, it is considered background.

2) Background Suppression: To suppress background responses in the detection map R, a three-stage suppression strategy is employed. First of all, an exponential mapping is applied to amplify the relative intensity differences between anomalous and background pixels. This operation enhances anomaly saliency by expanding the separation between weak anomaly signals and background clutter, thereby producing a refined detection map Y [as can be described in (23)]. Then, a guided filtering with spatial smoothing is applied, where the 2-D spatial structure of the HSI is leveraged through recursive filtering within a guided filter. This step enforces local spatial consistency, suppresses isolated noisy responses, and preserves structural boundaries, ensuring that anomalies are highlighted without introducing excessive blurring. Finally, a normalization operation is performed to standardize the refined detection scores across the entire image. This operation

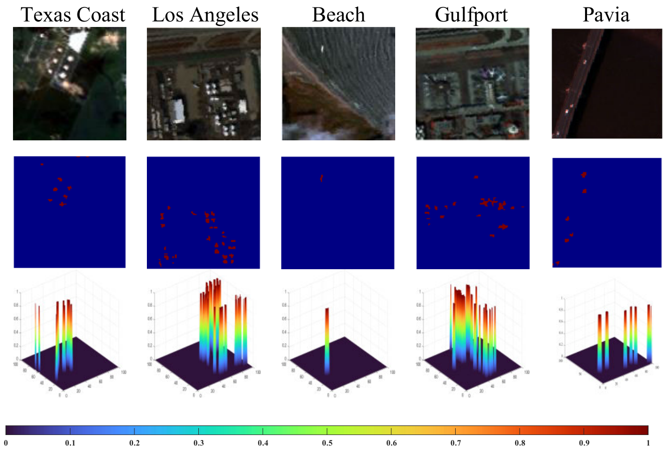

Fig. 4. (Top row) Pseudo-color images of five hyperspectral datasets, (middle row) corresponding ground truth maps, and (bottom row) 3-D view of the GT maps.

balances scale variations among different datasets and further clarifies the separation between anomalies and background regions. The overall suppression process is formulated in the following equations:

$$Y = \beta^R \tag{23}$$

$$h_i = \left(1 - \exp\left(-\frac{\sqrt{2}}{\varepsilon_1}\right)^o\right) y_i + \exp\left(-\frac{\sqrt{2}}{\varepsilon_1}\right)^o h_{i-1} \tag{24}$$

$$s_i = f_0 + \sum_{j=1}^{i}\left(1 + \frac{\varepsilon_1}{\varepsilon_2}|f_j - f_{j-1}|\right) \tag{25}$$

$$m_i = \frac{h_i - h_{\min}}{h_{\max} - h_{\min}} \tag{26}$$

where β is a parameter that controls the strength of background suppression, h_i denotes the filtered result, y_i represents the anomaly score obtained from the sliding dual-window, and m_i is the final detection result. The spatial distance between adjacent pixels in the transformed domain is denoted as $o = s_i - s_{i-1}$, while ε_1 and ε_2 correspond to the standard deviations in the spatial and range domains, respectively. Furthermore, f_i denotes the value of the ith pixel in the guidance image F, which is constructed as a pseudo-color image formed by the first three principal components (PCs) of the input HSI.

III. Experimental Results and Analysis

In this section, a comprehensive set of experiments is conducted on five hyperspectral datasets to validate the effectiveness of the proposed HTDC in terms of anomaly detection performance, where Section III-A describes the hyperspectral datasets used in the experiments, Section III-B outlines the comparison methods and evaluation metrics employed in the experiments, Section III-C provides a brief introduction of hardware environment and parameter settings, Section III-D presents the experimental results of the proposed method against eight state-of-the-art methods, Section III-E discusses the ablation studies, and Section III-F discusses the computational complexity analysis.

TABLE I
Some Hyperspectral Image Characteristics of Five Datasets

Datasets	Texas Coast	Los Angeles	Beach	Gulfport	Pavia
Number Bands	204	205	188	205	102
Spatial size	100×100	100×100	100×100	100×100	100×100
Spatial Resolution (m/pixel)	17.2	7.1	4.4	7.1	1.3
Anomaly pixels	67	232	11	170	71
Anomaly pixels percentage	0.67%	2.32%	0.11%	1.70%	0.71%
Sensor	AVIRIS	AVIRIS	AVIRIS	AVIRIS	ROSIS-3
Type of Anomaly	Buildings	Buildings	Unknown	Aircrafts	Vehicles

A. Experimental Datasets

To validate the proposed HTDC framework, experiments are conducted on five real-world datasets: Texas Coast, Los Angeles, Beach, Gulfport, and Pavia. These datasets represent diverse anomaly detection scenarios, with variations in spectral bands, spatial resolution, anomaly percentage, and anomaly type. The detailed characteristics of each dataset, including sensor type, spatial resolution, image size, number of spectral bands, number of anomalies, anomaly ratio, and anomaly types, are summarized in Table I. Furthermore, Fig. 4 presents the pseudo-color images, the corresponding ground-truth (GT) maps of anomalous targets, and the 3-D visualizations of the GT maps for each dataset. In Figs. 4–14, the color gradient from blue to red indicates anomaly probability, with regions approaching bright red signifying a higher likelihood of anomalies.

B. Comparison Methods and Evaluation Criteria

This section introduces comparison methods, including traditional methods and deep learning, as well as evaluation metrics, including AUC values, ROC curves, and box plots.

1) Comparison Methods: To validate the effectiveness of the proposed HTDC framework, it is compared against eight representative state-of-the-art approaches, encompassing both traditional statistical models and deep learning-based models. The traditional baselines include 2S-GLRT [56], GTVLRR [23], and AHMID [57], while the deep learning-based models include GAED [37], RGAE [35], Auto-AD [36], GIL-HAD [58], and NL2Net [59]. These baselines are carefully chosen to ensure a fair and comprehensive evaluation, as they collectively cover different methodological paradigms.

2) Evaluation Metrics: The detection performance of different methods is assessed using both visual and quantitative evaluation metrics. For visual evaluation, both 2-D view and 3-D view of detection results are provided, together with 3-D ROC (3-D ROC) curves and their corresponding 2-D projections (2-D ROC) of (P_D, P_F), (P_D, τ), and (P_F, τ) [60], [61]. These visualizations provide intuitive insight into the tradeoffs between detection probability, false-alarm rate, and threshold

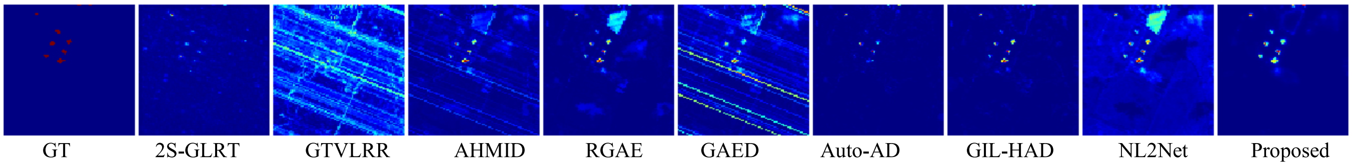

Fig. 5. 2-D view of anomaly detection results for the Texas Coast dataset.

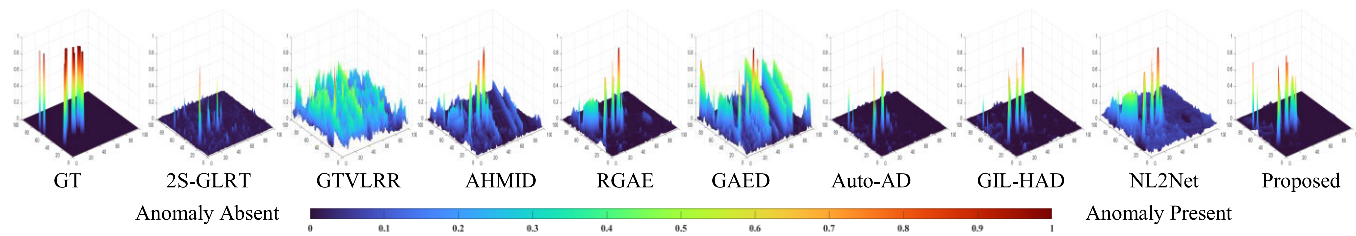

Fig. 6. 3-D view of anomaly detection results for the Texas Coast dataset.

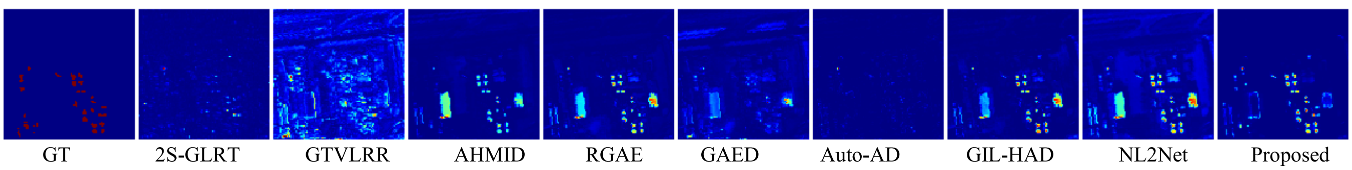

Fig. 7. 2-D view of anomaly detection results for the Los Angeles dataset.

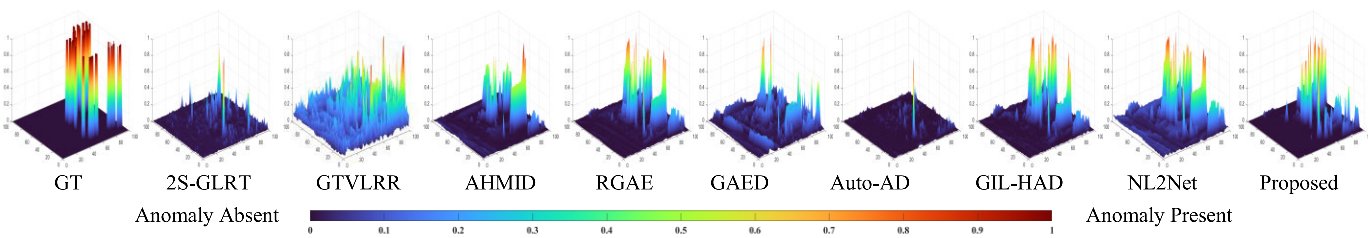

Fig. 8. 3-D view of anomaly detection results for the Los Angeles dataset.

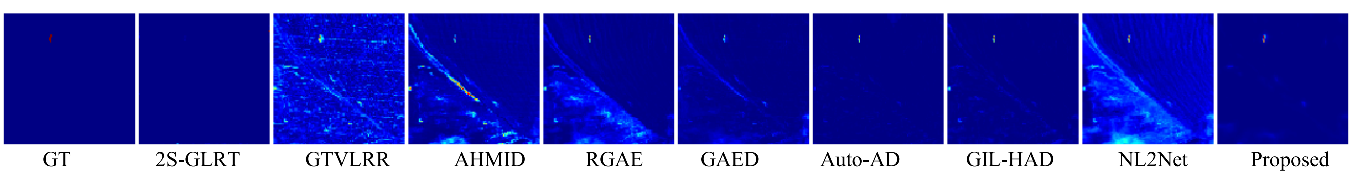

Fig. 9. 2-D view of anomaly detection results for the Beach dataset.

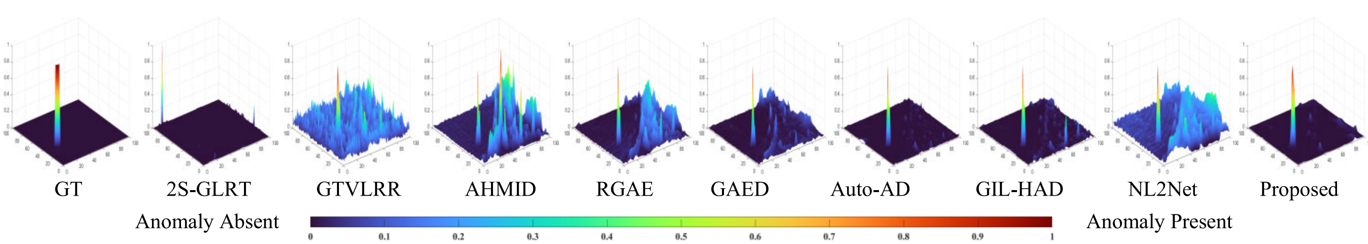

Fig. 10. 3-D view of anomaly detection results for the Beach dataset.

variation. For quantitative evaluation with a more rigorous comparison, multiple AUC metrics as listed in Table II are employed, including AUC (P_D, P_F), AUC_{TD}, AUC_{BS}, AUC_{ODP}, $\text{AUC}_{\text{TD-BS}}$, and AUC_{SNPR} [62]. AUC (P_D, P_F) is the fundamental metric, reflecting the overall effectiveness of the anomaly detector. AUC_{TD} evaluates the effectiveness of both the detector and the detectability of the targets. AUC_{BS} measures the background suppression capability of the detector.

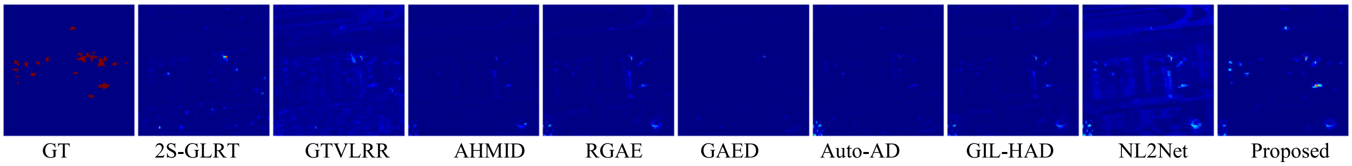

Fig. 11. 2-D view of anomaly detection results for the Gulfport dataset.

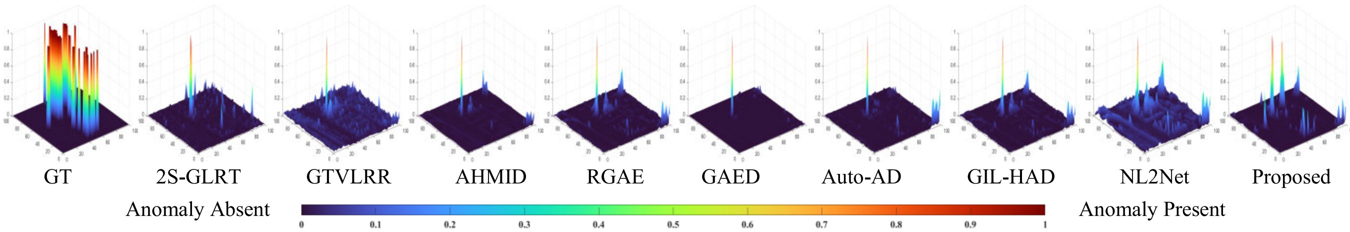

Fig. 12. 3-D view of anomaly detection results for the Gulfport dataset.

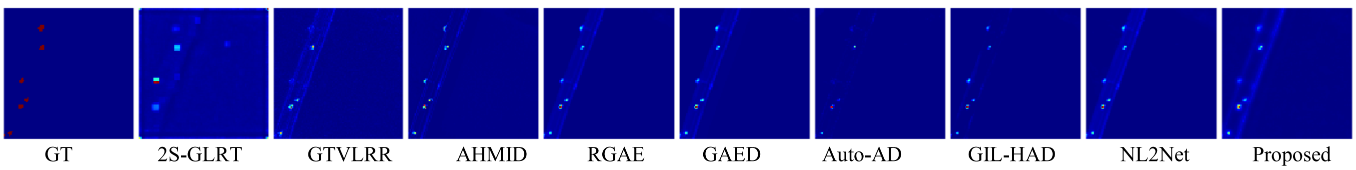

Fig. 13. 2-D view of anomaly detection results for the Pavia dataset.

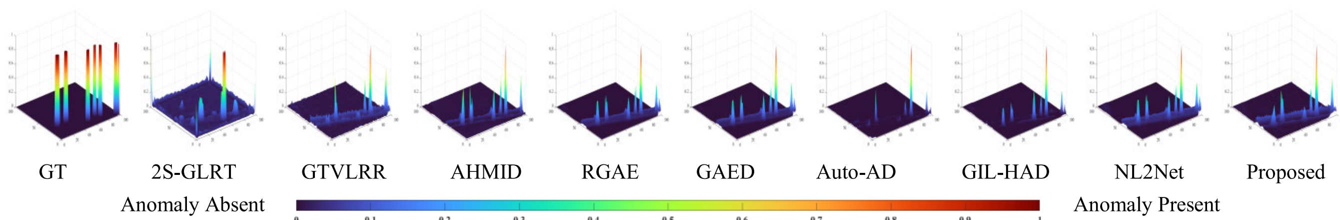

Fig. 14. 3-D view of anomaly detection results for the Pavia dataset.

TABLE II
SUMMARY OF THE AUC INDEXES

Index	Equation	Ideal value
$\mathrm{AUC}(P_D, P_F)$	$\mathrm{AUC}(P_D, P_F) = \int_{-\infty}^{+\infty} \mathrm{P_D}(\tau) d\mathrm{P_F}(\tau)$	1
$\mathrm{AUC}(P_D, \tau)$	$\mathrm{AUC}\ (P_D, \tau) = \int_{-\infty}^{+\infty} \mathrm{P_D}(\tau) d\tau$	1
$\mathrm{AUC}(P_F, \tau)$	$\mathrm{AUC}\ (P_F, \tau) = \int_{-\infty}^{+\infty} \mathrm{P_F}(\tau) d\tau$	0
$\mathrm{AUC_{TD}}$	$\mathrm{AUC_{TD}} = \mathrm{AUC}(P_D, P_F) + \mathrm{AUC}(P_D, \tau)$	2
$\mathrm{AUC_{BS}}$	$\mathrm{AUC_{BS}} = \mathrm{AUC}(P_D, P_F) - \mathrm{AUC}(P_F, \tau)$	1
$\mathrm{AUC_{ODP}}$	$\mathrm{AUC_{ODP}} = \mathrm{AUC}(P_D, P_F) + \mathrm{AUC}(P_D, \tau) - \mathrm{AUC}(P_F, \tau)$	2
$\mathrm{AUC_{TD\text{-}BS}}$	$\mathrm{AUC_{TDBS}} = \mathrm{AUC}(P_D, \tau) - \mathrm{AUC}(P_F, \tau)$	1
$\mathrm{AUC_{SNPR}}$	$\mathrm{AUC_{SNPR}} = \mathrm{AUC}(P_D, \tau) / \mathrm{AUC}(P_F, \tau)$	$+\infty$

$\mathrm{AUC_{ODP}}$ indicates the overall detection probability. $\mathrm{AUC_{TD\text{-}BS}}$ assesses the separability between background and target. $\mathrm{AUC_{SNPR}}$ provides a comprehensive assessment of detection performance, incorporating aspects such as signal-to-noise ratio and target–background discrimination. This combination of visual and quantitative metrics ensures a comprehensive and multiperspective evaluation of different models, allowing both qualitative interpretation and numerical comparison of detector performance.

Moreover, to further evaluate the separation between backgrounds and anomalies obtained by different algorithms, box plots are employed for comparative analysis. In particular, the values of the detected anomalies and backgrounds are first normalized, after which statistical computations are used to generate box representations for both categories. For all five datasets, values within the [10% and 90%] range are retained to characterize the distributions of anomaly and background pixels, reducing the influence of extreme outliers. It is worth noting that although the percentile thresholds can be adjusted, the [10%, 90%] range is commonly adopted in most cases.

C. Experimental Setup

This section mainly introduces the experimental environment and parameter configurations of both the proposed HTDC and the comparison baselines.

1) Experimental Environment: The experiments are conducted on two hardware platforms. For deep learning-based methods, the environment consists of a 24 vCPU AMD EPYC 7642 48-Core Processor and RTX 3090 24-GB GPU. The implementation is carried out using PyTorch 1.10, Python 3.8, and MATLAB R2023a. While for traditional statistical methods, an Intel Core i7-10510U CPU is used, with implementations completed in MATLAB R2023a.

2) Parameter Settings of HTDC: All hyperparameters of the proposed HTDC are empirically selected to ensure consistent performance across datasets: the OSBE parameters (k, s) are set to (9, 4) for all datasets. The hyperbolic Transformer module is configured with a depth of 2, 8 attention heads, and 8 neurons in each MLP layer. The sizes of the dual temporal queues (Q_1, Q_2) are set to (10 000, 2000) across all datasets. During the training phase, the network is optimized using the layerwise adaptive rate scaling (LARS) optimizer. The initial learning rate l is set to 0.05 and is dynamically adjusted using a cosine annealing scheduler. The temperature coefficients (τ_1, τ_2) are set to (0.1, 0.05), and the momentum coefficients (n, m) are set to (0.999, 0.8). The number of training epochs is set to 50 for the contrastive stage and 200 for dataset-specific finetuning. For all datasets, the batch size is set to 400, and the balance parameter λ is set to 0.5. The detection dual-window sizes are fixed at (3, 5). As for the background suppression process, β is set to 10, 20, 50, 20, and 10 for the Texas Coast, Los Angeles, Beach, Gulfport, and Pavia datasets, respectively. The parameter ε_1 (used to control the window size of the filter), the parameter ε_2 (used to control the ambiguity of the filter), and the number of iterations (performed in the guided filter) are set to (3, 2.5, 0.5), (3, 1.2, 0.5), (3, 2.5, 1), (3, 2.5, 0.5), and (3, 2.5, 0.5) for the Texas Coast, Los Angeles, Beach, Gulfport, and Pavia datasets, respectively.

3) Parameter Settings of Comparison Methods: For a fair comparison, baseline methods are configured according to the optimal parameters recommended in their respective references. For 2S-GLRT in [56], the outer and inner window sizes are set to 7 and 5. For GTVLRR in [23], the number of clusters K and total number of pixels per cluster P are set to 15 and 20, respectively, with parameters $\lambda = 0.5$, $\beta = 0.2$, and $\gamma = 0.05$. For AHMID [57], the parameters α, λ, and the number of layers are set to 1, 0.05, and 1, respectively. For GAED in [37], the second to sixth layer sizes are set to 100, 50, 25, 50, and 100, respectively, window size $c = 7$, learning rate $l = 0.4$, number of iterations = 300, and penalty $\beta = 1$. For RGAE in [35], the weight parameter, number of superpixels, and hidden layer dimension are set to 0.01, 150, and 100, respectively. For Auto-AD in [36], the number of training epochs is set to 50 across all five datasets, with a loss threshold of 1.0×10^{-5}. For GIL-HAD in [58], the size of the $N(p) = 7 \times 7$, learning rate $LR = 0.01$, reg noise = 0.1, layers = 5, and threshold $\tau = 0.000015$. For NL2Net in [59], the parameter PD = 5, the learning rate = 1.0×10^{-4}, and the number of training epochs = 100.

D. Results and Analysis

This section presents a comprehensive evaluation of the proposed HTDC method through a series of experiments, comparing it with eight state-of-the-art approaches mentioned above.

Figs. 5–14 show the detection maps of nine different anomaly detection methods on five HSI datasets. Since 2-D detection maps cannot fully reflect the detection performance, this article provides corresponding 3-D visualizations for enhanced interpretability. Figs. 15(b)–19(b) present the ROC curves in the (P_D, P_F) space for all methods on each dataset, where curves closer to the upper-left corner indicate better performance. Figs. 15(c)–19(c) illustrate the ROC curves in the (P_D, τ) space, where curves approaching the upper-left corner signify superior results. Figs. 15(d)–19(c) display the ROC curves in the (P_F, τ) space, with curves closer to the lower-left corner indicating better performance. Figs. 15(a)–19(a) show the 3-D ROC curves for each dataset, from which eight key evaluation metrics are derived to comprehensively assess detection capability. The highest performance indices are highlighted in bold, and the second-best results are marked with an underline.

To provide a detailed analysis of the evaluation metrics, the Texas Coast dataset is taken as an example. As shown in Fig. 5, the detection results of different methods are compared. Traditional methods fail to achieve clear and complete anomaly detection. Except for GAED and Auto-AD, most deep learning-based methods can identify the contours and locations of anomalies; however, they exhibit relatively weak background suppression. Taking GIL-HAD as an example, both the proposed HTDC and GIL-HAD can effectively detect anomalies, but the targets detected by GIL-HAD are less prominent compared to those by HTDC. As reported in Table III, the proposed method achieves a detection accuracy of 0.9987, which is 12% higher than that of GTVLRR, the method with the lowest performance. For the AUC (P_F, τ) metric, the proposed method ranks second, slightly behind Auto-AD. However, HTDC outperforms Auto-AD by 1.12% in the AUC (P_D, P_F) metric, indicating that Auto-AD's superior anomaly detection comes at the cost of weaker background suppression. Compared with GIL-HAD, the HTDC improves the overall detection accuracy by 0.54% and outperforms it across the remaining seven evaluation metrics.

For the Los Angeles dataset, the detection results are shown in Fig. 7. AHMID, RGAE, GIL-HAD, NL2Net, and HTDC can accurately localize all anomaly positions and contours. However, the results of 2S-GLRT and Auto-AD contain significant background interference, embedding the detected targets into the background and making them difficult to observe. GTVLRR is severely affected by noise. As shown in Table III, with background suppression evaluated using the AUC (P_F, τ), the proposed method ranks second. For all other evaluation metrics, HTDC outperforms all competing methods. Compared with Auto-AD, although the proposed method ranks slightly lower in background suppression, it surpasses Auto-AD across all other metrics.

For the Beach dataset, the detection results are shown in Fig. 9. Except for 2S-GLRT, all other methods are able to accurately localize the anomaly positions and contours. The 2S-GLRT result map contains significant background interference, embedding the detected targets and making them

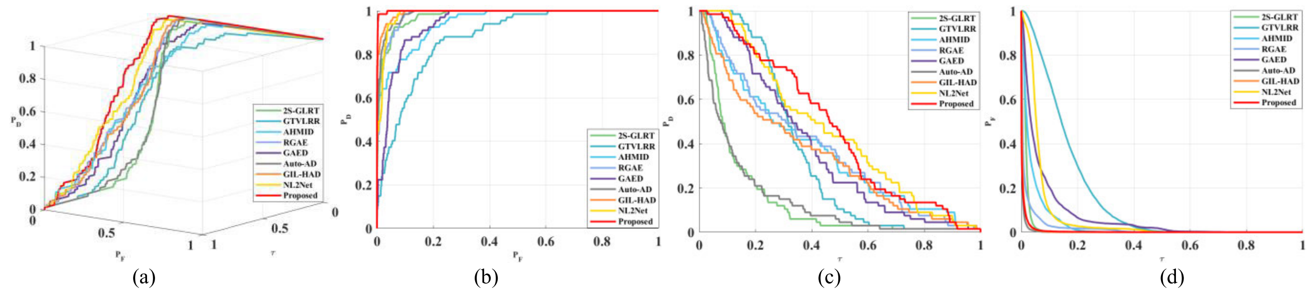

Fig. 15. ROC curves for the Texas Coast dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (P_D, P_F). (c) 2-D ROC curve of (P_D, τ). (d) 2-D ROC curve of (P_F, τ).

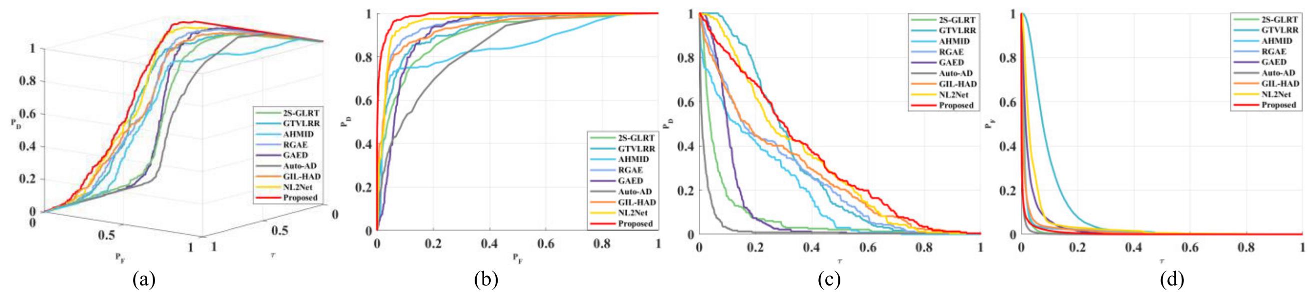

Fig. 16. ROC curves for the Los Angeles dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (P_D, P_F). (c) 2-D ROC curve of (P_D, τ). (d) 2-D ROC curve of (P_F, τ).

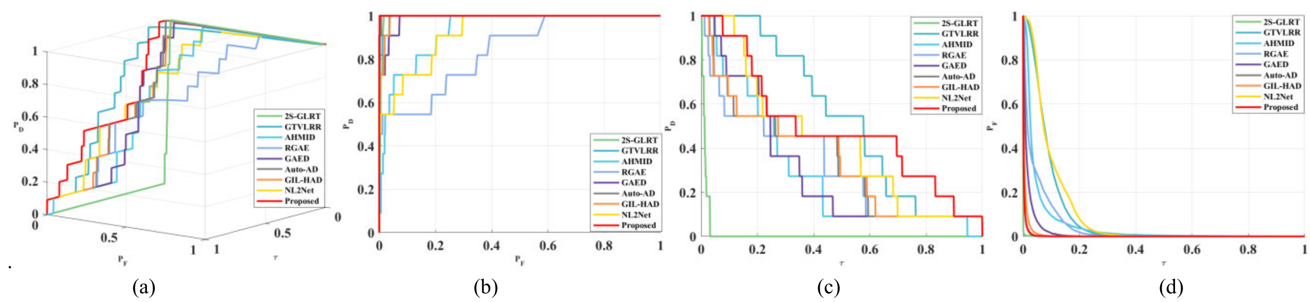

Fig. 17. ROC curves for the Beach dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (P_D, P_F). (c) 2-D ROC curve of (P_D, τ). (d) 2-D ROC curve of (P_F, τ).

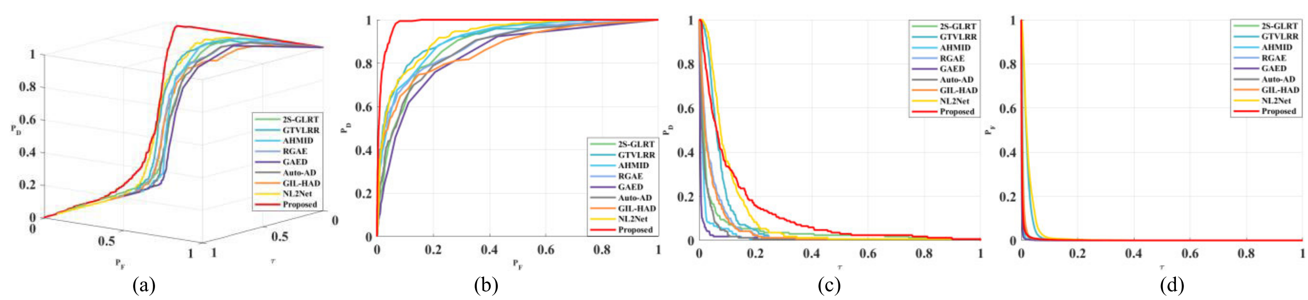

Fig. 18. ROC curves for the Gulfport dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (P_D, P_F). (c) 2-D ROC curve of (P_D, τ). (d) 2-D ROC curve of (P_F, τ).

unobservable, along with a number of false alarms. GTVLRR is heavily affected by noise. AHMID, RGAE, GAED, GIL-HAD, and NL2Net exhibit varying degrees of background interference, and AHMID in particular produces a large number of false detections. As shown in Table III, the HTDC method ranks second in terms of AUC (P_F, τ), AUC (P_D, τ), and AUC_{TD}, while outperforming all other methods in the remaining evaluation metrics. Compared with Auto-AD, although the detection map of HTDC shows similar background suppression and target detection performance, the proposed method outperforms Auto-AD across all evaluation metrics, as detailed in Table III.

For the Gulfport dataset, the detection results are shown in Fig. 11. HTDC and NL2Net successfully detect most small

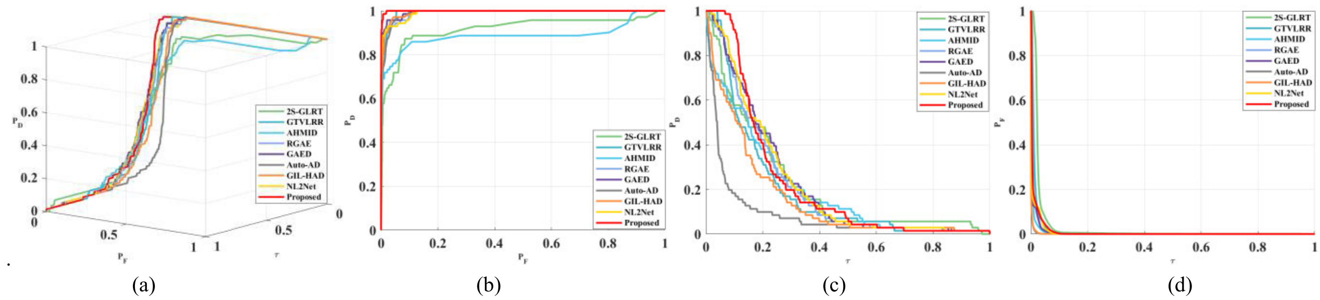

Fig. 19. ROC curves for the Pavia dataset. (a) 3-D ROC curve. (b) 2-D ROC curve of (P_D, P_F). (c) 2-D ROC curve of (P_D, τ). (d) 2-D ROC curve of (P_F, τ).

TABLE III

AUC Values Computation of Different Methods on Five Datasets: (I) Texas Coast, (II) Los Angeles, (III) Beach, (IV) Gulfport, and (V) Pavia

Dataset	AUC	2S-GLRT	GTVLRR	RGAE	GAED	Auto-AD	AHMID	GIL-HAD	NL2Net	Proposed
I	AUC (P_D, P_F) ↑	0.9788	0.8787	0.9807	0.9439	0.9875	0.9483	0.9933	0.9849	0.9987
	AUC (P_D, τ) ↑	0.1417	0.3440	0.3690	0.3727	0.1430	0.3638	0.3387	0.4559	0.4647
	AUC (P_F, τ) ↓	0.0178	0.1601	0.0178	0.0695	0.0039	0.0371	0.0058	0.0653	0.0052
	AUC_{TD} ↑	1.1205	1.2227	1.3498	1.3165	1.1305	1.3122	1.3320	1.4408	1.4635
	AUC_{BS} ↑	0.9610	0.7186	0.9630	0.8743	0.9836	0.9112	0.9875	0.9197	0.9936
	AUC_{ODP} ↑	1.1027	1.0627	1.3320	1.2470	1.1266	1.2751	1.3262	1.3756	1.4583
	$\text{AUC}_{\text{TD-BS}}$ ↑	0.1239	0.1840	0.3513	0.3032	0.1391	0.3268	0.3330	0.3906	0.4595
	AUC_{SNPR} ↑	7.9733	2.1492	20.7775	5.3597	36.6666	9.8129	58.5780	6.9859	89.7629
II	AUC (P_D, P_F) ↑	0.8985	0.9257	0.9489	0.9175	0.8436	0.8497	0.9341	0.9625	0.9887
	AUC (P_D, τ) ↑	0.0786	0.3125	0.2494	0.1134	0.0269	0.1884	0.2679	0.3279	0.3433
	AUC (P_F, τ) ↓	0.0140	0.0968	0.0200	0.0307	0.0043	0.0193	0.0181	0.0455	0.0073
	AUC_{TD} ↑	0.9771	1.2382	1.1982	1.0308	0.8706	1.0381	1.2020	1.2904	1.3321
	AUC_{BS} ↑	0.8845	0.8289	0.9289	0.8868	0.8393	0.8304	0.9160	0.9169	0.9814
	AUC_{ODP} ↑	0.9631	1.1414	1.1783	1.0002	0.8663	1.0188	1.1839	1.2448	1.3247
	$\text{AUC}_{\text{TD-BS}}$ ↑	0.0646	0.2157	0.2294	0.0827	0.0226	0.1691	0.2498	0.2824	0.3360
	AUC_{SNPR} ↑	5.6111	3.2284	12.4801	3.6974	6.2450	9.7832	14.8152	7.2004	46.8634
III	AUC (P_D, P_F) ↑	0.9942	0.9952	0.8414	0.9884	0.9989	0.9331	0.9931	0.9252	0.9999
	AUC (P_D, τ) ↑	0.0138	0.5368	0.3160	0.3015	0.3400	0.2907	03455	0.4290	0.4860
	AUC (P_F, τ) ↓	0.0008	0.0879	0.0423	0.0162	0.0033	0.0479	0.0062	0.0966	0.0032
	AUC_{TD} ↑	1.0079	1.5320	1.1573	1.2899	1.3388	1.2238	1.3386	1.3541	1.4859
	AUC_{BS} ↑	0.9934	0.9073	0.7991	0.9723	0.9955	0.8851	0.9869	0.8286	0.9967
	AUC_{ODP} ↑	1.0071	1.4441	1.1150	1.2738	1.3355	1.1758	1.3324	1.2575	1.4827
	$\text{AUC}_{\text{TD-BS}}$ ↑	0.0130	0.4489	0.2737	0.2853	0.3386	0.2428	0.3393	0.3323	0.4828
	AUC_{SNPR} ↑	17.0594	6.1073	7.4702	18.6372	101.5022	6.0645	55.6488	4.4396	151.2471
IV	AUC (P_D, P_F) ↑	0.8906	0.9156	0.8872	0.8436	0.8691	0.9093	0.8638	0.9243	0.9860
	AUC (P_D, τ) ↑	0.0498	0.0786	0.0479	0.0133	0.0268	0.0245	0.0447	0.1029	0.1187
	AUC (P_F, τ) ↓	0.0071	0.0211	0.0056	0.0016	0.0036	0.0046	0.0057	0.0257	0.0039
	AUC_{TD} ↑	0.9404	0.9942	0.9351	0.8569	0.8959	0.9338	0.9085	1.0273	1.1048
	AUC_{BS} ↑	0.8835	0.8946	0.8816	0.8420	0.8654	0.9046	0.8581	0.8986	0.9821
	AUC_{ODP} ↑	0.9333	0.9732	0.9295	0.8553	0.8923	0.9291	0.9028	1.0015	1.1009
	$\text{AUC}_{\text{TD-BS}}$ ↑	0.0428	0.0575	0.0423	0.0117	0.0232	0.0199	0.0390	0.0772	0.1148
	AUC_{SNPR} ↑	7.0471	3.7281	8.5536	8.3125	7.3928	5.2785	7.8345	4.0003	30.5079
V	AUC (P_D, P_F) ↑	0.9185	0.9927	0.9927	0.9948	0.9910	0.8891	0.9933	0.9920	0.9993
	AUC (P_D, τ) ↑	0.2143	0.1881	0.2070	0.2223	0.0878	0.1955	0.1550	0.2204	0.2235
	AUC (P_F, τ) ↓	0.0277	0.0138	0.0048	0.0056	0.0015	0.0079	0.0016	0.0074	0.0097
	AUC_{TD} ↑	1.1328	1.1808	1.1997	1.2171	1.0697	1.0846	1.1483	1.2124	1.2230
	AUC_{BS} ↑	0.8908	0.9789	0.9879	0.9892	0.9895	0.8812	0.9917	0.9846	0.9896
	AUC_{ODP} ↑	1.1051	1.1670	1.1949	1.2115	1.0682	1.0767	1.1467	1.2050	1.2133
	$\text{AUC}_{\text{TD-BS}}$ ↑	0.1866	0.1743	0.2022	0.2167	0.0863	0.1876	0.1534	0.2130	0.2138
	AUC_{SNPR} ↑	7.7365	13.6304	43.1250	39.6964	58.5333	24.7468	96.8750	29.7838	23.0412

*The optimal performances are marked in red, while the sub-optimal performances are marked in blue.

targets as well as some large ones. In contrast, the other methods fail to detect small targets by visual inspection, as they tend to over-suppress the background at the cost of missing a large portion of the anomalies. The targets detected by GAED remain embedded in the background and are barely visible upon inspection. As illustrated in Fig. 11, HTDC demonstrates superior anomaly highlighting and background suppression capabilities compared with other competing algorithms. According to Table III, although the AUC (P_F, τ) metric is not dominant, the proposed method ranks first across the remaining seven evaluation metrics, indicating strong performance in anomaly detection, background suppression, and

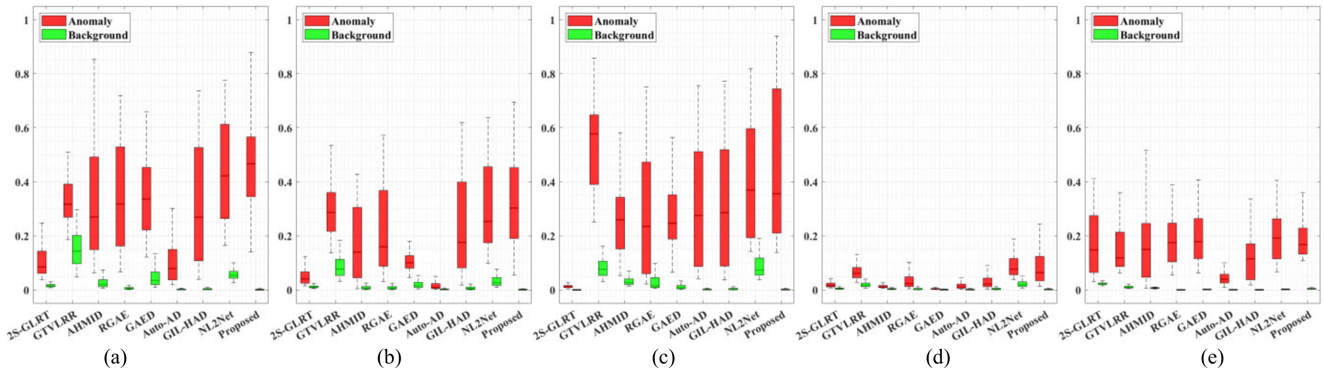

Fig. 20. Box plot of comparison algorithms for five datasets. (a) Coast dataset. (b) Los Angeles dataset. (c) Beach dataset. (d) Gulfport dataset. (e) Pavia.

background–target separation. Moreover, the method achieves a detection accuracy of 0.9860, which is 6.17% higher than the second-best result achieved by NL2Net. Compared with GIL-HAD, the proposed method improves the detection rate by 12.22%.

For the Pavia dataset, the detection results are shown in Fig. 13. The experimental results demonstrate the superior robustness of the proposed method compared to state-of-the-art competitors. While Auto-AD struggles significantly with a lower detection rate of 0.9910, likely due to over-suppression of targets in complex backgrounds, the proposed method maintains high confidence and stability. According to the table, although the AUC_{SNPR} metric is not dominant compared to Auto-AD, the proposed method ranks first in critical metrics, including AUC_{TD}, AUC_{ODP}, and AUC (P_D, τ). This indicates exceptional capabilities in background-target separation and anomaly highlighting without sacrificing signal integrity. In particular, the method achieves a remarkable detection accuracy of 0.9993. This result is 0.45% higher than the runner-up GAED and outperforms Auto-AD by a significant margin of 0.83%, comprehensively validating its effectiveness in handling challenging scenarios.

To further validate the detection results, 3-D views of the detection results were performed, as shown in Figs. 6, 8, 10, 12, and 14. Compared to traditional 2-D detection results, the 3-D visualization represents the "anomaly score" in the form of height, offering a clearer representation of the prominence of anomalies relative to the background. As shown in Figs. 6, 8, 10, 12, and 14, anomalies in all five datasets are significantly elevated above the background, demonstrating the method's strong separability. Moreover, the anomaly detection results exhibit high spatial consistency with the ground truth maps, being more closely aligned with the true anomalies than other methods, thereby reflecting a higher detection accuracy. Simultaneously, the background regions in all five datasets are relatively flat, indicating strong background suppression capability. In conclusion, the proposed method effectively enhances detection performance and the ability to separate anomalies from the background.

To visually verify the separability between the background and anomalies, boxplots of the detection scores from nine methods are illustrated in Fig. 20. To mitigate the influence of outliers, the distributions are plotted after excluding the top and bottom 10% of pixel values. In each subfigure, red boxes indicate the distribution of anomalous targets, while green boxes represent the background. The central horizontal line denotes the median, and the upper and lower boundaries indicate the value range. Ideally, a compact background box (indicating strong suppression) and a large gap between the target and background boxes (indicating high separability) are desired. As observed, on the Texas Coast dataset, the proposed HTDC exhibits strong target responses with background maximum values compressed near zero. For the Beach and Gulfport datasets, although the median target scores of HTDC are slightly lower than those of GTVLRR and NL2Net, the latter methods exhibit significantly higher background residuals (elevated green boxes). Consequently, HTDC achieves a superior contrast ratio. Consistently across all five datasets, HTDC yields the distinctest margin between the target and background distributions with virtually no overlap. The highly compact background boxes further confirm the method's robust suppression capability. In summary, the proposed method effectively suppresses background interference and maximizes the separability of anomalies.

Overall, the proposed HTDC method successfully and comprehensively detects all of the anomalous targets while demonstrating strong background suppression capabilities in complex environments. As shown in Figs. 15–19, the 3-D ROC and the corresponding 2-D ROC curves further highlight the superior overall performance of HTDC. In addition, the eight metrics in Table III also indicate that HTDC exhibits excellent detection capabilities across various aspects. Therefore, HTDC shows outstanding detection performance in both qualitative and quantitative analyses.

E. Ablation Article

In this section, a comprehensive ablation study is conducted to evaluate the effectiveness of different modules within the proposed HTDC method. To clearly isolate the contribution of each component, the study is divided into two parts: 1) an analysis of the momentum CL framework and 2) an analysis of the core functional components, specifically the geometric space and data augmentation strategies. The visualization of the detection accuracy for these variants is presented in Figs. 21 and 22.

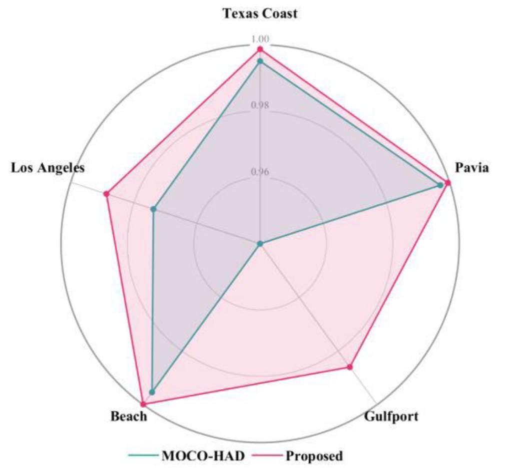

Fig. 21. Detection accuracy comparison between MOCO-HAD and the proposed method.

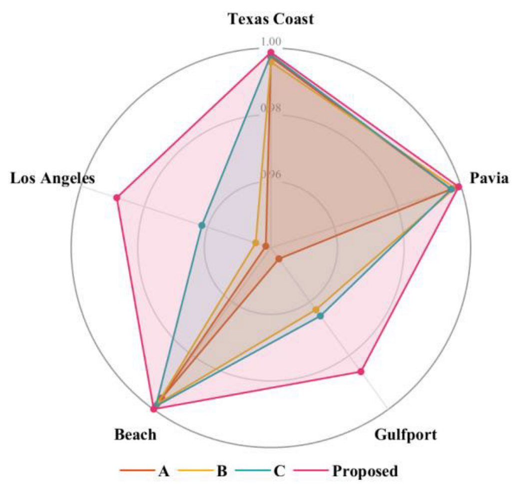

Fig. 22. Detection accuracy comparison for variants of the proposed method.

1) Effectiveness of Parameter Update Strategy: The impact of the training framework is first evaluated by comparing the proposed DMCL with the conventional single-momentum [momentum contrast (MoCo)] approach (referred to as MOCO-HAD) [52], [63]. For MOCO-HAD, data augmentation is performed using Gaussian noise (with $\beta = 0.005$), the momentum coefficient is set to 0.999, the temperature parameter is set to 0.07, and the queue length is set to 10 000. Crucially, to ensure a fair comparison, the feature extraction networks of both the proposed method and MCT are consistently based on the spectral feature extraction via hyperbolic Transformer proposed in this article.

Fig. 21 visualizes the AUC (P_D, P_F) performance comparison using a radar chart. Visually, the curve of the proposed method completely encloses that of MOCO-HAD, indicating superior detection accuracy across all five hyperspectral datasets. A critical observation lies in the shape of the polygons. While MOCO-HAD exhibits a sharp contraction along the Gulfport axis, the proposed method maintains a high AUC (P_D, P_F) of 0.9860. This result confirms that our method effectively eliminates the "performance bottleneck" observed in the baseline, demonstrating exceptional capability in handling challenging scenarios.

TABLE IV
DETAILED SETTINGS OF DIFFERENT CASES IN ABLATION EXPERIMENTS

Module	A	B	C	Proposed
Multi-view	×	×	√	√
Single-view (Gaussian Noise)	√	×	×	×
Single-view (Spectral Absence)	×	√	×	×
Euclidean Transformer	×	×	√	×
Hyperbolic Transformer	√	√	×	√

*Incluted block: √, Unincluted block: ×.

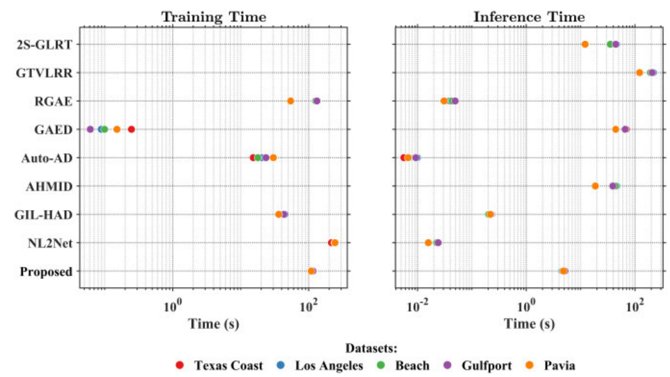

Fig. 23. Running time of the comparative methods on five datasets.

2) Contribution of Core Components: Based on the dual-momentum framework, the specific contributions of the hyperbolic geometry and multiview augmentation were further investigated. As detailed in Table IV, four variants (denoted as A, B, C, and proposed) are designed to isolate these components. In the experimental setup for Case C, the Euclidean Transformer (i.e., the standard Transformer architecture) is configured with a depth of 2, 8 attention heads, and 8 neurons in each MLP layer. These hyperparameters are kept consistent with the Hyperbolic setting to ensure a fair comparison regarding the geometric space.

Fig. 22 presents the radar chart of AUC (P_D, P_F) performance across five hyperspectral datasets. Visually, the enclosed area of the proposed method is the largest, indicating superior overall performance. More importantly, the shape consistency reveals significant differences in robustness. Methods A and B exhibit a "star-like" shape with sharp contractions along the Los Angeles and Gulfport axes, indicating their vulnerability to complex background clutter in these specific scenes. In contrast, the proposed method maintains a quasi-regular pentagonal shape, demonstrating that it effectively mitigates the performance degradation observed in other methods. The proposed method essentially "closes the gap" on the challenging datasets while retaining near-perfect scores on easier tasks (e.g., Texas Coast), confirming its excellent generalization capability.

F. Computational Complexity Analysis

To comprehensively evaluate the proposed HTDC, the computational efficiency of all methods is further compared. Table V reports the computational complexity metrics, including the number of parameters (Params) and floating point operations (FLOPs) on the five datasets. Fig. 23 displays

TABLE V
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS ON FIVE DATASETS

Dataset	Metrics	RGAE	GAED	Auto-AD	GIL-HAD	NL2Net	Proposed
Texas Coast	Params (M)	0.0411	0.0538	3.2487	3.4472	0.7698	1.4604
	FLOPs (G)	0.8162	1.0675	11.9589	12.9496	15.3421	0.0399
Los Angeles	Params (M)	0.0413	0.0540	3.2501	3.4487	0.7699	1.4604
	FLOPS (G)	0.8200	1.0715	11.9698	12.9605	15.3440	0.0407
Beach	Params (M)	0.0379	0.0506	3.2262	3.4247	0.7682	1.4604
	FLOPS (G)	0.7520	1.0035	11.7848	12.7755	15.3114	0.0367
Gulfport	Params (M)	0.0413	0.0540	3.2501	3.4487	0.7699	1.4604
	FLOPS (G)	0.8200	1.0715	11.9698	12.9605	15.3440	0.0407
Pavia	Params (M)	0.0206	0.0333	3.1050	3.3035	0.7599	1.4604
	FLOPs (G)	0.4080	0.6596	10.8491	11.8398	15.1462	0.0200

*The optimal performances are marked in red, while the sub-optimals are marked in blue.

the running times (e.g., training time and inference time) of various detectors for all detectors applied to the five datasets under consideration.

1) Model Complexity: Regarding model complexity, HTDC demonstrates an exceptional lightweight advantage. As reported in Table V, the parameter count of HTDC is constant at 1.4604 M, which is significantly lower than that of large-scale models like Auto-AD. Most notably, HTDC achieves the lowest computational cost in terms of FLOPs. In particular on the Texas Coast dataset, HTDC requires only 0.0399 G FLOPs, standing in sharp contrast to the 11.9589 G required by Auto-AD. This indicates that our method is extremely energy-efficient, making it particularly suitable for resource-constrained platforms where power consumption is a critical constraint.

2) Time Consumption: The computational efficiency of the compared methods is visualized in Fig. 23. Note that a logarithmic scale is used to accommodate the vast differences in time costs. In the train phase, traditional methods (e.g., GTVLRR and 2S-GLRT) do not involve a training stage. In contrast, deep learning-based methods, including the proposed HTDC, require a training phase to learn feature representations. The proposed HTDC incurs a moderate training time, which is comparable to that of RGAE but higher than that of simple lightweight models like Auto-AD. Crucially, it is worth noting that training is a one-time, offline computational cost. Once the model is trained, it can be deployed permanently without reoptimization. Therefore, the training overhead does not impact the real-time detection capability. Real-time capability is determined by inference time. Traditional methods suffer from heavy computational burdens due to iterative optimization, taking tens to hundreds of seconds per image. In contrast, HTDC demonstrates a significant speedup. Taking the Texas Coast dataset as an example, HTDC records an inference time of approximately 4.5 s, achieving a speedup of nearly two orders of magnitude compared to traditional approaches. While slightly slower than millisecond-level networks due to the sophisticated feature extraction, this acceptable latency is a worthy tradeoff for the superior detection accuracy.

IV. CONCLUSION

To address the vulnerability of reconstruction-based HAD methods to contamination from anomalous pixels during background reconstruction, this article proposes a multiview DMCL framework. The proposed method adopts a two-stage design. In the training stage, a multiview data augmentation strategy is employed to generate semantically consistent yet diverse sample pairs for CL. Subsequently, a hyperbolic Transformer-based feature extraction module is integrated with CL, incorporating both a dual-momentum update strategy and hyperbolic space optimization to facilitate multilevel modeling of local geometric structures and global spectral information. This design enhances the model's ability to learn discriminative spectral representations for distinguishing anomalies from the background. In the testing stage, the pretrained model is applied to anomaly detection using a sliding dual-window strategy, enabling fine-grained response to anomalous regions. To further optimize detection results, several post-processing techniques are employed, including exponential transformation, normalization, and guided filtering, effectively suppressing background interference and enhancing anomaly saliency. Extensive experiments and ablation studies conducted on five real-world hyperspectral datasets confirm that the proposed HTDC framework consistently outperforms eight state-of-the-art methods in terms of detection accuracy and robustness.

Future efforts will be directed toward extending the framework to address large-scale anomaly detection scenarios through implementation optimization. Moreover, spatial–spectral attention mechanisms are planned to be integrated to enhance performance, with potential applications on resource-constrained edge devices.

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